

A FERMION-BOSON COMPOSITE MODEL OF QUARKS AND LEPTONS

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Quark and lepton masses and flavor-mixing angles are estimated on the basis of a fermion-boson composite model where the (u, d), (c, s) and (t, b) quarks are assigned to the diagonal elements π^8 , η^8 and η^1 , respectively, in $3 \times 3^* = 8 + 1$ of the SU(3)-generation symmetry.

Recently there has been considerable interest in composite models of quarks and leptons. One of the major motivations for studying these models is the generation problem. There have been many legitimate studies [1] of the generation problem, where solutions for generation numbers have been sought under some field-theoretical constraints (the 't Hooft condition [2], and so on). Besides such an orthodox study, it may also be important to seek a model which provides anyhow the observed mass spectra and mixing patterns of quarks and leptons, even if it is highly speculative.

The author [3] has recently proposed a new formula for the Cabibbo angle θ_c ,

$$\tan \theta_c = \sqrt{3} \frac{\sqrt{m_\mu} - \sqrt{m_e}}{2\sqrt{m_\tau} - \sqrt{m_\mu} - \sqrt{m_e}} = 0.225, \quad (1)$$

which is in excellent agreement with experiment. In the model, there are three subquarks h_i ($i = 1, 2, 3$), which carry the generation quantum number, and a lepton contains two h , so that three generations of leptons are understood as 3^* in $3 \times 3 = 3^* + 6$ of the SU(3)-generation symmetry, while a quark contains h_i and $\bar{h}^i$, so that three generations of quarks are explained as the three diagonal elements π^8 , η^8 and η^1 of $3 \times 3^*$. The existence of a small SU(3)_G-symmetry breaking induces for leptons merely mass differences, while for quarks it causes π^8 - η^8 mixing together with mass differences.

Of course, we need not take our π^8 and η^8 literal-

ly as the diagonal elements of $3 \times 3^*$ bosons ($h\bar{h}$). In the derivation of eq. (1) it is essential to provide the following mass matrices [4] for leptons (ℓ_1, ℓ_2, ℓ_3) and quarks (q_1, q_2, q_3):

$$M^\ell = \begin{pmatrix} \delta_1^\ell & 0 & 0 \\ 0 & \delta_2^\ell & 0 \\ 0 & 0 & \delta_3^\ell \end{pmatrix}, \quad (2)$$

$$M^q = \begin{pmatrix} \delta_1^q & 0 & 0 \\ 0 & \delta_2^q & 0 \\ 0 & 0 & \delta_3^q \end{pmatrix} + \Delta^q \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad (3)$$

where the SU(3)_G-symmetry breaking parameters δ_i satisfy:

$$\delta_1^\nu \approx \delta_2^\nu \approx \delta_3^\nu \approx 0, \quad \delta_1^u \approx \delta_2^u \ll \delta_3^u, \quad (4)$$

$$\delta_i^e \propto (z_i + z_0)^2, \quad \delta_i^d \propto z_i^2, \quad (5)$$

$$z_1 + z_2 + z_3 = 0, \quad (6)$$

and $e_1 = e^-$, $e_2 = \mu^-$ and $e_3 = \tau^-$. In the limit of $\delta_1^q = \delta_2^q = \delta_3^q \equiv \delta^q$, the mass eigenstates of quarks become $\pi^8 \equiv (q_1 - q_2)/\sqrt{2}$, $\eta^8 \equiv (q_1 + q_2 - 2q_3)/\sqrt{6}$ and $\eta^1 \equiv (q_1 + q_2 + q_3)/\sqrt{3}$, whose masses are δ^q , δ^q and $\delta^q + 3\Delta^q$, respectively. The existence of a small SU(3)_G-symmetry breaking $\delta_3^q - (\delta_2^q + \delta_1^q)/2 (> 0)$ makes the η^8 state massive relative to the π^8 state, and $\delta_1^q - \delta_2^q \neq 0$ induces π^8 - η^8 mixing:

$$\tan 2\theta^q \approx \sqrt{3} \frac{\delta_1^q - \delta_2^q}{2\delta_3^q - \delta_2^q - \delta_1^q} \left(1 + \frac{2\delta_3^q - \delta_2^q - \delta_1^q}{6\Delta^q} \right) \quad (7)$$

under the approximation $\delta_i^q \ll \Delta^q$. For "down" quarks d_i , eq. (7) with relation (5) leads to

$$\tan \theta^d \approx \sqrt{3} \frac{z_2 - z_1}{2z_3 - z_2 - z_1} = \sqrt{3} \frac{\sqrt{m_\mu} - \sqrt{m_e}}{2\sqrt{m_\tau} - \sqrt{m_\mu} - \sqrt{m_e}}, \quad (8)$$

where we have used relation (6). Therefore we can obtain formula (1) for the Cabibbo angle $\theta_c = \theta^d - \theta^u$, because $\theta^u \approx 0$ under the assumption (4).

A clue to the generation puzzle may be provided by the matrices (2) and (3) with relations (4), (5) and (6).

In ref. [3] relation (4) is, a priori, an assumption. It is not self-evident whether a model which provides relation (5) can satisfy relation (4) or not.

In this paper, as a specific example of a model⁺¹ which provides the mass matrices (2) and (3), we propose a composite model where quarks and leptons are fermion-boson two-body bound systems made of some fundamental constituents (preons), and thereby we estimate quark and lepton masses and the Kobayashi-Maskawa mixing angles [6].

In our model, there are three families. Each family is constructed of a color-triplet fermion $F_{ai} = (U_{ai}, D_{ai})$, a color-triplet boson C_{ai} and a color-singlet boson S_i , where a and i are $SU(3)_C$ - and $SU(3)_G$ -indices, respectively⁺². The preons F_{ai} , C_{ai} and S_i are assign-

ed to N of $SU(N)$ -subcolor and therefore they are confined inside subcolor-singlet states. The quantum numbers of the preons are listed in table 1. We assume the $SU(2)_L \times SU(2)_R \times U(1)_{B-L}$ gauge model [8] for the electroweak interactions. We also assume that $SU(3)_C$ and $SU(N)_{sc}$ are gauge groups, but we consider that $SU(3)_G$ is not gauged.

In addition to the $SU(N)$ -subcolor confinement force, we assume a confinement force which interacts only between preons with the same family index i . If the superstrong interaction is scalar, and the coupling constants of C_{ai} and S_i have the same sign, but that of F_{ai} has opposite sign relative to those, then the subcolor-singlet systems $(F_{ai}\bar{C}^{aj})$ and $(F_{ai}\bar{S}^j)$ have a negative potential term $-\delta_{ij}V_s$, while the other two-body systems $(F_{ai}\bar{F}^{aj})$, $(C_{ai}\bar{C}^{aj})$, $(S_i\bar{S}^j)$ and $(C_{ai}\bar{S}^j)$ have a positive potential term $+\delta_{ij}V_s$. We assume that only the simultaneous existence of the two confinement potentials $-\delta_{ij}V_s$ (with $i=j$) and $[(N^2-1)/2N]V_{sc}$ makes the two-body systems massless. Generally, subcolor-singlet states which are made of N -body preons ($N \geq 3$) are massive because of the appearance of the potential terms $+\delta_{ij}V_s$. After all, we obtain only two massless states

$$\ell_i = \frac{1}{\sqrt{3}} \sum_a (F_{ai}\bar{C}^{ai}) \quad \text{and} \quad q_{ai} = (F_{ai}\bar{S}^i). \quad (9)$$

(Color-octet leptons are forbidden because of color confinement.)

We assume that the Δ^q -term in the mass matrix (3) is provided by $U(3)_G$ -invariant interactions of preons F_{ai} and S_i with a hypothetical $U(3)_G$ -nonet boson $\phi_G^{(9)i}$. The boson $\phi_G^{(9)i}$ does not interact with the preon C_{ai} .

In order to provide the $SU(3)_G$ -symmetry breaking mass terms δ_i^q and δ_i^l , we suppose an interaction

⁺¹ Other specific examples of such a model can be found in refs. [4] and [5].

⁺² Although our model is like the Fritzsch-Mandelbaum model [7], our preon S_i has no color in contrast to their model, where the color-triplet quark is made of $3^* \times 3^*$ of $SU(3)_C$.

Table 1
Quantum numbers of preons. For the "flavor charges" Q_F , see text.

preons	spin	$SU(N)_{sc}$	$SU(3)_C$	B	L	Q_E	Q_F
U_{ai}	1/2	N	3	0	0	1/2	z_0
D_{ai}	1/2	N	3	0	0	-1/2	$-z_i$
C_{ai}	0	N	3	0	-1	1/2	z_0
S_i	0	N	1	-1/3	0	-1/6	$-\sqrt{3}z_i$

$$g_F \sum_i \frac{z_i}{(z_1^2 + z_2^2 + z_3^2)^{1/2}} \times \left(\frac{1}{\sqrt{3}} \sum_a \bar{D}^{ai} \gamma_\mu D_{ai} + S^{*i} \delta_\mu^{\rightarrow} S_i \right) \phi_F^{(8)\mu}, \quad (10)$$

where $\phi_F^{(8)}$ behaves as if it were a member of the $SU(3)_G$ -octet (this may be seen as analogous to the fact that the photon behaves as if it were a member of the old-flavor $SU(3)$ -octet). The other preons U_{ai} and C_{ai} interact with another hypothetical vector boson $\phi_F^{(1)}$, which is $SU(3)_G$ -singlet, as follows:

$$g_F \frac{1}{\sqrt{3}} \sum_i \left(\frac{1}{\sqrt{3}} \sum_a \bar{U}^{ai} \gamma_\mu U_{ai} + \frac{1}{\sqrt{3}} \sum_a C^{*ai} \delta_\mu^{\rightarrow} C_{ai} \right) \phi_F^{(1)\mu}. \quad (11)$$

If the bosons ($\phi_F^{(1)}$, $\phi_F^{(8)}$) have the mass matrix

$$\mu_F \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad (12)$$

then the mass eigenstates are $\phi_F \equiv (\phi_F^{(1)} - \phi_F^{(8)})/\sqrt{2}$ and $\phi'_F \equiv (\phi_F^{(1)} + \phi_F^{(8)})/\sqrt{2}$, which have the masses 0 and $2\mu_F$, respectively. We assume that μ_F is sufficiently large, so that contributions of ϕ'_F to the mass matrices (2) and (3) are negligibly small. We call the coupling constants of preons with ϕ_F "flavor charges" Q_F [4]. The flavor charges of preons are as follows:

$$Q_F(U_{ai}) = z_0, \quad Q_F(C_{ai}) = z_0, \\ Q_F(D_{ai}) = -z_i, \quad Q_F(S_i) = -\sqrt{3} z_i, \quad (13)$$

in units of $g_F/\sqrt{6} z_0$, where z_0 is defined by

$$z_0 = (z_1^2 + z_2^2 + z_3^2)^{1/2}/\sqrt{3}. \quad (14)$$

We assume that the $SU(3)_G$ -symmetry breaking mass term $\delta_i^{(AB)}$ for the system which is made of preons A and B is given by the sum of the potential term $Q_F(A)Q_F(B)v_F$ and the self-energy terms $\frac{1}{2}[Q_F(A)]^2 v_F + \frac{1}{2}[Q_F(B)]^2 v_F$, that is,

$$\delta_i^{(AB)} = \frac{1}{2}[Q_F(A) + Q_F(B)]^2 v_F, \quad (15)$$

where v_F is a parameter which depends on the "effective radius" r_0 of the system (for example, such a parameter as $v_F(r_0) = 1/4\pi r_0$ if the potential is Coulomb-like). Here we have assumed that the self-energy terms of the preons A and B are characterized by the same parameter $v_F(r_0)$ as in the potential term

of the preons A and B. From the assumption (15), we obtain the $SU(3)_G$ -symmetry breaking mass terms δ_i as follows:

$$\delta_i^\nu = 0, \quad \delta_i^u = (\sqrt{3} z_i + z_0)^2, \\ \delta_i^e = (z_i + z_0)^2, \quad \delta_i^d = (\sqrt{3} - 1)^2 z_i^2, \quad (16)$$

where a common factor $(g_F/\sqrt{6} z_0)(v_F/2)$ has been omitted.

We can obviously get the prediction

$$m_{\nu_e} = m_{\nu_\mu} = m_{\nu_\tau} = 0.$$

Note that relation (14) leads to a sum rule ⁺³:

$$m_e + m_\mu + m_\tau = \frac{2}{3}(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2. \quad (17)$$

From the observed values of m_e and m_μ , we can predict

$$m_\tau = 1.777 \text{ GeV}, \quad (18)$$

which is in good agreement with the observed value [9] $m_\tau = 1.784 \pm 0.004 \text{ GeV}$.

From eqs. (7) and (16), under the approximation $\Delta^q \rightarrow \infty$, we can successfully obtain

$$\theta^u = 0.014 \text{ (0.80°)}, \quad (19)$$

$$m_u = 0.080 \text{ GeV}, \quad m_c = 2.434 \text{ GeV}, \\ m_d = 0.084 \text{ GeV}, \quad m_s = 0.252 \text{ GeV}, \quad (20)$$

together with relation (8), although the predicted value of m_c is somewhat large. Note that the model (16) favorably leads to $m_u \approx m_d$.

Since F_{ai} and $\bar{C}^{ai}$ are bounded by the color-confinement potential $-\frac{4}{3} V_c(r_0)$ in addition to the confinement potential

$$- \left(V_s(r_0) + \frac{N^2 - 1}{2N} V_{sc}(r_0) \right),$$

it is likely that the effective radius of a lepton is small compared to that of a quark, that is, $r_0^l < r_0^q$, and therefore that $v_F(r_0^l) > v_F(r_0^q)$. For example, if we assume that $v_F(r_0) \propto r_0^{-1}$, $V_c(r_0) = V_{sc}(r_0) \propto r_0$ and $V_s(r_0) = \text{constant}$, we obtain

⁺³ The sum rule (17) [and therefore the prediction (18)] was first guessed on the basis of another specific model [4].

$$\frac{v_F(r_0^q)}{v_F(r_0^e)} = \frac{r_0^q}{r_0^e} = \left(\frac{N^2 - 1}{2N} \right) / \left(\frac{N^2 - 1}{2N} + \frac{4}{3} \right). \quad (21)$$

$N = 4, 5$ and 6 lead to $v_F^q/v_F^e = 45/77, 9/14$ and $35/51$, respectively, and therefore to $m_c = 1.423$ GeV, 1.565 GeV and 1.671 GeV, respectively. Of course, in our model, the assumption (21) is not essential. Another suppression mechanism for the predicted quark masses may be considered. However, as we need the explicit value of the suppression factor when we discuss the mixing effect with the η^1 state (t and b quarks), we will use hereafter the value $v_F^q/v_F^e \approx 0.643$ ($=9/14$) as a provisional value which provides a phenomenologically favorable value of m_c . The choice $v_F^q/v_F^e \approx 0.643$ gives the modified values

$$\begin{aligned} m_u &= 0.051 \text{ GeV}, & m_c &= 1.565 \text{ GeV}, \\ m_d &= 0.054 \text{ GeV}, & m_s &= 0.162 \text{ GeV}, \end{aligned} \quad (22)$$

instead of the prediction (20).

The predicted value (19) of θ^u is favorably small. Moreover, the small deviation from eq. (1), which is caused by $\theta^u \neq 0$, is improved by consideration of the small mixing effect with the η^1 state:

$$\begin{aligned} \tan 2\theta_c &\approx \tan 2\theta^d - \tan 2\theta^u \\ &\approx \sqrt{3} \left(\frac{\delta_1^d - \delta_2^d}{2\delta_3^d - \delta_2^d - \delta_1^d} + \frac{\delta_1^d - \delta_2^d}{6\Delta^d} - \frac{\delta_1^u - \delta_2^u}{2\delta_3^u - \delta_2^u - \delta_1^u} \right) \\ &= 0.476 + 0.024 - 0.028, \end{aligned} \quad (23)$$

where we have assumed that m_t is large enough to neglect the term $(\delta_1^u - \delta_2^u)/\Delta^u$ and we have used $m_b \approx 4.7$ GeV and $v_F^q/v_F^e \approx 0.643$, that is,

$$3\Delta^d \approx m_b - (\delta_1^d + \delta_2^d + \delta_3^d)/3 \approx 4.6 \text{ GeV}.$$

The new estimate (23) scarcely modifies the prediction (1):

$$\tan \theta_c = 0.224 \quad (\theta_c = 12.6^\circ). \quad (24)$$

Of the Kobayashi–Maskawa mixing matrix elements are of greatest interest to us those related to b -decays. The models based on the mass matrix (3), in general, predict

$$\begin{aligned} U_{b \rightarrow u} &\approx \lambda^d \tan 2\theta_c, \\ U_{b \rightarrow c} &\approx -\lambda^d (1 - \lambda^u/\lambda^d), \end{aligned} \quad (25)$$

under the approximation $\delta_i^q \ll \Delta^q$, where

$$\lambda^q = (2\delta_3^q - \delta_2^q - \delta_1^q)/9\sqrt{2}\Delta^q. \quad (26)$$

If we suppose that m_t is large compared to m_b , we can predict

$$(U_{b \rightarrow u}/U_{b \rightarrow c})^2 \approx \tan^2 2\theta_c, \quad (27)$$

which is consistent with the recent experimental results [10], $\Gamma(b \rightarrow u) \ll \Gamma(b \rightarrow c)$, from B -meson decays. Note that if it is experimentally established that $\Gamma(b \rightarrow u)/\Gamma(b \rightarrow c) \approx 0$ and m_t is not so large compared to m_b , for example, $m_t \approx 20$ GeV, then not only our specific model but also all models based on the mass matrix (3) are ruled out because of a sizable value of λ^u/λ^d in eq. (25).

In our specific model, $U_{b \rightarrow c}$ is estimated as

$$U_{b \rightarrow c} \approx -\lambda^d \approx -0.023, \quad (28)$$

under the approximation $\lambda^d/\lambda^u \approx 0$. The predicted value (28) does not conflict with the constraint $|U_{b \rightarrow c}| > 0.01$ [11], which is deduced from the upper limit of the B -meson lifetimes.

In conclusion, we have proposed a specific model which provides the mass matrices (2) and (3) with the relations (4), (5) and (6). Using only the observed values of m_e and m_μ , we have predicted the other lepton masses and quark masses except for m_b and m_t . (Unfortunately, in our model, m_t and m_b contain the free parameters Δ^u and Δ^d , which are undecided by m_e and m_μ .) It is worth noting the prediction $m_\tau = 1.777$ GeV. Since the predicted value of m_c is somewhat large, we must assume a suppression mechanism for the predicted quark masses, which may be related to a difference of the “scales” of quarks and leptons. Our predicted Kobayashi–Maskawa mixing matrix does not conflict with the experiments so far.

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